Chapter 1. Sets

Exercise 1.1. Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$. Find
$$A \cap B, A \cup B, A \setminus B, B \setminus A, A \Delta B.$$

Exercise 1.2. Let $U = \{1, 2, \ldots, 10\}$ be the universal set. If $A = \{2, 4, 6, 8, 10\}$ and $B = \{1, 3, 5, 7, 9\}$, then find
$$A^c, B^c, (A \cup B)^c, (A \cap B)^c.$$

Exercise 1.3. Write down the power set of each of the following sets:
$$\{1, 2\}, \{1, 2, 3\}, \{1, 2, 3, 4\}.$$

Exercise 1.4. Identify the set
$$\left\{x \in \mathbb{R} \setminus \{0\} : x + \frac{1}{x} \geq 2\right\}.$$

Exercise 1.5. Let A, B be subsets of $\mathbb{R}$ defined by
$$A = \{x \in \mathbb{R} : x^2 \geq 0\},$$
$$B = \{x \in \mathbb{R} : x^3 \geq 0\}.$$
Determine the sets $A \cup B$, $A \cap B$, $A \setminus B$, $B \setminus A$.

Exercise 1.6. For a positive integer k, let A_k denote the set of integral multiples of k, that is,
$$A_k := \{r \in \mathbb{Z} : r = k\ell \text{ for some } \ell \in \mathbb{Z}\}.$$
Let m, n be positive integers. For each of the following statements, determine the equivalent conditions on the integers m, n.

1. $A_m \subseteq A_n$
2. $A_m \subsetneq A_n$
3. $A_m \not\subseteq A_n$
4. $A_m = A_n$

Chapter 2. Induction

Exercise 2.1. Show that
$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$
for all $n \in \mathbb{N}$.

Exercise 2.2. Show that
$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$
for all $n \in \mathbb{N}$.

Exercise 2.3. Show that

$$\frac{1}{1\cdot 2\cdot 3}+\frac{1}{2\cdot 3\cdot 4}+\cdots+\frac{1}{n(n+1)(n+2)}=\frac{n(n+3)}{4(n+1)(n+2)}$$

for all $n \in \mathbb{N}$.

Exercise 2.4. Show that

$$3+11+\cdots+(8n-5)=4n^2-n$$

for all $n \in \mathbb{N}$.

Exercise 2.5. Show that

$$1^2+3^2+\cdots+(2n-1)^2=\frac{4n^3-n}{3}$$

for all $n \in \mathbb{N}$.

Exercise 2.6. Show that

$$1^2-2^2+3^2-4^2+\cdots+(-1)^{n+1}n^2=(-1)^{n+1}\frac{n(n+1)}{2}$$

for all $n \in \mathbb{N}$.

Exercise 2.7. Show that

$$1^2+2^2+3^2+\cdots+n^2=\frac{n(n+1)(2n+1)}{6},$$

$$1^3+2^3+3^3+\cdots+n^3=\left(\frac{n(n+1)}{2}\right)^2$$

hold for any positive integer n.

Exercise 2.8. Show that

$$\frac{1}{n+1}+\frac{1}{n+2}+\cdots+\frac{1}{2n}=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots+\frac{1}{2n-1}-\frac{1}{2n}$$

for all $n \in \mathbb{N}$.

Exercise 2.9. Show that n^3+5n is divisible by 6 for all $n \in \mathbb{N}$.

Exercise 2.10. Show that $5^{2n}-1$ is divisible by 8 for all $n \in \mathbb{N}$.

Exercise 2.11. Show that 5^n-4n-1 is divisible by 16 for all $n \in \mathbb{N}$.

Exercise 2.12. Show that 6^n-5n-1 is divisible by 25 for all $n \in \mathbb{N}$.

Exercise 2.13. Show that $n^3+(n+1)^3+(n+2)^3$ is divisible by 9 for all $n \in \mathbb{N}$.

Exercise 2.14. Show that

$$\frac{1}{\sqrt{1}}+\frac{1}{\sqrt{2}}+\cdots+\frac{1}{\sqrt{n}}>\sqrt{n}$$

for all $n \in \mathbb{N}$ satisfying $n > 1$.

Chapter 3. Matrices

Exercise 3.1. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 1 \\ 1 & 6 \end{pmatrix}$. Compute $2A + 3B$, A^2B, AB^3, A^T.

Exercise 3.2. Suppose A is a 3×2 matrix, and

$$A\begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 8 \\ 14 \end{pmatrix},$$

$$A\begin{pmatrix} 3 \\ 7 \end{pmatrix} = \begin{pmatrix} -4 \\ 8 \\ 20 \end{pmatrix}.$$

Determine

$$A\begin{pmatrix} 103 \\ 407 \end{pmatrix}, A\begin{pmatrix} 97 \\ 393 \end{pmatrix}.$$

Exercise 3.3. Show that every vector in $\mathbb{R}^2$ can be expressed as a linear combination of the vectors

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix}.$$

Exercise 3.4. Write the following system of equations in matrix form $Ax = b$.

$$2x + 3y = 5,$$

$$4x - y = 1.$$

Determine if A is invertible. If it is, find A^{-1} and use it to solve the system.

Exercise 3.5. If A, B are 2×2 matrices, then show that

$$\det(AB) = \det(A)\det(B).$$

Exercise 3.6. Determine which of the following matrices are upper-triangular, lower-triangular, or neither.

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}, \begin{pmatrix} 7 & 8 \\ 0 & 9 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{pmatrix}, \begin{pmatrix} 7 & 0 \\ 8 & 9 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \\ 5 & 0 & 6 \end{pmatrix}.$$

Exercise 3.7. Determine which of the following matrices are symmetric, which are skew-symmetric, and which are neither.

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}, \begin{pmatrix} 7 & 8 \\ 8 & 9 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix},$$

$$\begin{pmatrix} 0 & 2 & 3 \\ -2 & 0 & 5 \\ -3 & -5 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 8 \\ -8 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Exercise 3.8. Determine which of the following matrices are orthogonal.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Exercise 3.9. Let

$$A = \begin{pmatrix} 1 & -5 & 6 \\ -4 & 8 & -9 \\ 7 & 2 & 10 \end{pmatrix}.$$

Find $\det(A)$ and $\operatorname{adj}(A)$.

Exercise 3.10. Let

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 3 & 2 & 1 \\ 4 & 1 & 2 \end{pmatrix}.$$

Find $\det(A)$ and $\operatorname{adj}(A)$.

Exercise 3.11. Let θ be a real number satisfying $0 < \theta < \frac{\pi}{2}$. Show that the matrix

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

is orthogonal.

Exercise 3.12. Let A be an orthogonal matrix. Show that A is invertible and that

$$A^{-1} = A^T.$$

Exercise 3.13. Write the following system of equations in matrix form $Ax = b$.

$$\begin{aligned} x_1 + 2x_2 - x_3 + 4x_4 &= 5, \\ 2x_1 + 3x_2 + x_3 + 5x_4 &= 8, \\ -x_1 + 4x_2 - 2x_3 - 3x_4 &= -4, \\ 3x_1 - x_2 + 4x_3 + 2x_4 &= 7. \end{aligned}$$

Perform Gaussian elimination on the augmented matrix $(A \mid b)$, to reduct it to row echelon form. Using this form, determine if the system has a solution. If it does, find all solutions of the above system in the variables x_1, x_2, x_3, x_4.

Exercise 3.14. Perform row reduction on the matrix

$$\begin{pmatrix} 1 & 2 & -5 \\ 2 & 5 & -7 \\ -1 & -2 & 4 \end{pmatrix}$$

to reduce it to row echelon form. Using these reductions, find the inverse of the above matrix, if it exists.

Exercise 3.15. Transform the matrix

$$\begin{pmatrix} 0 & 2 & -3 & 0 & 0 \\ 1 & -1 & 4 & 0 & 5 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 3 & -5 & 0 & 1 \\ 0 & 3 & -5 & -1 & 2 \end{pmatrix}$$

into a matrix in row echelon form using elementary row operations.

Exercise 3.16. Solve the system of equations in the variables x_1, x_2, x_3, x_4.

$$\begin{aligned} 2x_2 - 3x_3 &= 0, \\ x_1 - x_2 + 4x_3 &= 5, \\ x_4 &= -2, \\ 3x_2 - 5x_3 &= 1, \\ 3x_2 - 5x_3 - x_4 &= 2. \end{aligned}$$

Exercise 3.17. Let

$$A = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix},$$

where a, b, c are scalars. Show that $\det(A) = abc$, and that

$$\operatorname{adj}(A) = \begin{pmatrix} bc & 0 & 0 \\ 0 & ca & 0 \\ 0 & 0 & ab \end{pmatrix}.$$

Exercise 3.18. Compute the determinant of the matrix

$$A = \begin{pmatrix} 1 & -5 & 6 \\ 2 & 0 & 9 \\ -1 & 2 & -1 \end{pmatrix}$$

to determine whether it is invertible. If A is invertible, then find its inverse using its adjoint.

Exercise 3.19. Use Gaussian elimination to determine the row echelon form of the matrix

$$A = \begin{pmatrix} 1 & -5 & 6 \\ 2 & 0 & 9 \\ -1 & 2 & -1 \end{pmatrix}.$$

Use the row echelon form to determine whether A is invertible. If A is invertible, then find its inverse using elementary matrices, corresponding to the elementary row operations used to transform A into its row echelon form.

Exercise 3.20. Solve the following system of equations in the variables x, y, z.

$$\begin{aligned} x + 2y + 3z &= 1, \\ 4x - 5y + 6z &= 2, \\ 7x - y + 10z &= 3. \end{aligned}$$

Chapter 4. The space $\mathbb{R}^n$

Exercise 4.1. Let S be a subset of $\mathbb{R}^n$, containing precisely one element v. Show that S is linearly independent if and only if $v \neq 0$.

Exercise 4.2. In each of the following, determine whether the given vectors are linearly independent or dependent. Justify your answers.

1. $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$, and $\begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$ in $\mathbb{R}^3$.
2. $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ in $\mathbb{R}^3$.
3. $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ in $\mathbb{R}^2$.
4. $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ in $\mathbb{R}^2$.
5. $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$, and $\begin{pmatrix} 7 \\ 8 \\ 10 \end{pmatrix}$ in $\mathbb{R}^3$.

Exercise 4.3. Let $v_1 = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$, $v_2 = \begin{pmatrix} 4 \\ 0 \\ -1 \end{pmatrix}$, $v_3 = \begin{pmatrix} -1 \\ 5 \\ 2 \end{pmatrix}$. Determine whether the vectors v_1, v_2, v_3 are linearly independent or dependent. Justify your answer.

Exercise 4.4. Determine whether the vectors $\begin{pmatrix} 1 \\ 2 \\ -1 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 4 \\ 0 \\ 5 \\ -2 \end{pmatrix}$, $\begin{pmatrix} -2 \\ 1 \\ 3 \\ 4 \end{pmatrix}$, and $\begin{pmatrix} 3 \\ -1 \\ 2 \\ 1 \end{pmatrix}$ in $\mathbb{R}^4$ are linearly independent or dependent. Justify your answer.

Exercise 4.5. Show that the standard basis vectors

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \ldots, e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

of $\mathbb{R}^n$ are linearly independent.

Exercise 4.6. Show that any finite subset of $\mathbb{R}^n$ containing the zero vector is linearly dependent.

Exercise 4.7. Let V denote the set of all vectors of $\mathbb{R}^n$ whose first component is equal to zero, that is,

$$V = \left\{ \begin{pmatrix} 0 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} : x_i \in \mathbb{R} \text{ for all } i = 2, 3, \ldots, n \right\}.$$

Show that V is a linear subspace of $\mathbb{R}^n$.

Exercise 4.8. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}$ is a linear subspace of $\mathbb{R}^3$.

Exercise 4.9. Let (a, b, c) be a nonzero element of $\mathbb{R}^3$. Show that the set

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : ax + by + cz = 0 \right\}$$

is a linear subspace of $\mathbb{R}^3$.

Exercise 4.10. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x \geq 0, y \geq 0 \right\}$ is not a linear subspace of $\mathbb{R}^2$.

Exercise 4.11. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x = 1 \right\}$ is not a linear subspace of $\mathbb{R}^2$.

Exercise 4.12. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x = y \right\}$ is a linear subspace of $\mathbb{R}^2$.

Exercise 4.13. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : z = 0 \right\}$ is a linear subspace of $\mathbb{R}^3$.

Exercise 4.14. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 1 \right\}$ is not a linear subspace of $\mathbb{R}^3$.

Exercise 4.15. Let A be an $m \times n$ matrix. Let V be the set of solutions to the system of equations

$$Au = 0,$$

that is,

$$V = \{v \in \mathbb{R}^n : Av = 0\}.$$

Show that V is a linear subspace of $\mathbb{R}^n$.

Exercise 4.16. Let A be an $m \times n$ matrix. Let V be the set of vectors which can be expressed as a linear combination of the columns of A, that is,

$$V = \{v \in \mathbb{R}^m : v = Au \text{ for some } u \in \mathbb{R}^n\}.$$

Show that V is a linear subspace of $\mathbb{R}^m$.

Exercise 4.17. Let U, V be the subsets of $\mathbb{R}^3$ defined as

$$U = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + 2y = 0 \right\},$$

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}.$$

Show that both U and V are linear subspaces of $\mathbb{R}^3$. Show that $U \cap V$ is a linear subspace of $\mathbb{R}^3$. Determine a basis for the intersection $U \cap V$ of U and V.

Exercise 4.18. Let V be a linear subspace of $\mathbb{R}^n$. Show that $V + V = V$.

Exercise 4.19. Let U, V be the subsets of $\mathbb{R}^3$ defined as

$$U = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + 2y = 0 \right\},$$

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}.$$

Determine the sum of U and V.

Exercise 4.20. Provide examples of linear subspaces V_1, V_2, V_3 of $\mathbb{R}^3$ satisfying

$$\{0\} \subsetneq V_1 \subsetneq V_2 \subsetneq V_3 \subseteq \mathbb{R}^3.$$

Exercise 4.21. Let e_1, e_2 denote the standard basis vectors of $\mathbb{R}^2$, that is,

$$e_1 := \begin{pmatrix} 1 \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Show that each of the following sets of vectors generates $\mathbb{R}^2$:

1. $\{e_1, e_2\}$,
2. $\{e_1, e_1 + e_2\}$,
3. $\{e_2, e_1 + e_2\}$,
4. $\{e_1 - e_2, e_1 + e_2\}$,
5. $\{e_1, e_2, e_1 + e_2\}$.

Prove that

$$\{ae_1 + be_2, ce_1 + de_2\}$$

generates $\mathbb{R}^2$ if and only if $ad - bc \neq 0$.

Exercise 4.22. Determine a generating set for the subspace

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}.$$

Exercise 4.23. Show that $\mathbb{R}^n$ is generated by the standard basis vectors $\{e_1, e_2, ..., e_n\}$, where

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, ..., e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

Exercise 4.24. Show that the standard basis vectors

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, ..., e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

form a basis of $\mathbb{R}^n$.

Exercise 4.25. Determine whether the subsets of $\mathbb{R}^2$ as in Exercise 4.21 are bases of $\mathbb{R}^2$.

Exercise 4.26. Determine a basis of the linear subspace

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x - 2y + 6z = 0 \right\}.$$

of $\mathbb{R}^3$.

Exercise 4.27. Determine a basis of the linear subspace

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y = 0 \right\}.$$

of $\mathbb{R}^3$.

Exercise 4.28. Determine a basis of each of the subspaces as in the prior examples or exercises.

Exercise 4.29. Show that $\dim(\mathbb{R}^n) = n$.

Exercise 4.30. Determine a basis and the dimension of each of the subspaces as in the prior examples or exercises.

Exercise 4.31. Determine a basis and the dimension of the null space of

$$A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{pmatrix}.$$

Exercise 4.32. Determine a basis and the dimension of the column space of

$$A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{pmatrix}.$$

Chapter 5. Functions

Exercise 5.1. Consider the function $f : (0,1) \to \mathbb{R}$ defined by

$$f(x) = \frac{1}{x} \text{ for all } x \in (0,1).$$

Show that f is injective but not surjective.

Exercise 5.2. Let $M_n(\mathbb{R})$ denote the set of all $n \times n$ matrices with real entries. Show that the function $\det : M_3(\mathbb{R}) \to \mathbb{R}$, defined by

$$A \mapsto \det(A) \text{ for all } A \in M_3(\mathbb{R}),$$

is not injective, but surjective.

Exercise 5.3. Let $\mathrm{GL}_2(\mathbb{R})$ denote the set of all invertible 2×2 matrices with real entries. Show that the function $\det : \mathrm{GL}_2(\mathbb{R}) \to \mathbb{R} \setminus \{0\}$ defined by

$$A \mapsto \det(A) \text{ for all } A \in \mathrm{GL}_2(\mathbb{R})$$

is surjective but not injective.

Exercise 5.4. Show that the map

$$f : \{..., -6, -4, -2, 0, 2, 4, 6, ...\} \to \{..., -3, -2, -1, 0, 1, 2, 3, ...\}$$

defined by

$$f(n) = \frac{n}{2} \text{ for all } n \in \{..., -6, -4, -2, 0, 2, 4, 6, ...\}$$

is bijection between the set of all even integers and the set of all integers.

Exercise 5.5. Show that the map

$$f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$$

defined by

$$f(m, n) = 2^{m-1}(2n - 1) \text{ for all } (m, n) \in \mathbb{N} \times \mathbb{N}$$

is a bijection between $\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$.

Exercise 5.6. Let a, b be real numbers satisfying $a < b$. Show that the map

$$f : (0, 1) \to (a, b)$$

defined by

$$f(x) = a + (b - a)x \text{ for all } x \in (0, 1)$$

is a bijection.

Using the above or otherwise, show that for any real numbers a, b, c, d satisfying $a < b$ and $c < d$, there is a bijection between the intervals (a, b) and (c, d).

Exercise 5.7. The function

$$| \cdot | : \mathbb{R} \to \mathbb{R}$$

defined by

$$|x| := \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0 \end{cases}$$

is called the **absolute value** function. Show that it is neither injective nor surjective.

Exercise 5.8. Consider the function

$$f : \mathbb{R} \to (0, 1]$$

defined by

$$f(x) = \frac{1}{1 + |x|} \text{ for all } x \in \mathbb{R}.$$

Show that f is not injective. Determine $\mathrm{im}(f)$.

Exercise 5.9. The map

$$\| \cdot \| : \mathbb{R}^2 \to \mathbb{R},$$

defined by

$$\|(x, y)\| := \sqrt{x^2 + y^2} \text{ for all } (x, y) \in \mathbb{R}^2,$$

is called the **norm** on $\mathbb{R}^2$. Show that it is neither injective nor surjective.

Exercise 5.10. Determine the inverse image of the set $\{1\}$ under the norm map

$$\|\cdot\| : \mathbb{R}^2 \to \mathbb{R}.$$